



SARDAR PATEL UNIVERSITY

B.C.A. (Semester- III) (FOR BATCH 2011-2017)

US03CBCA02: Object Oriented Programming and C++

Date: 23-09-2022

Time: 12.30 pm – 02.30 pm

Marks: 70

Q-1

Select the correct option from the following questions.

(10)

- 1) _____ means "To bind data & functions into a single unit".
 (A) Inheritance (B) Encapsulation
 (C) Polymorphism (D) Data Hiding
- 2) _____ is known as scope resolution operator.
 (A) & (B) >>
 (C) << (D) ::
- 3) In _____ data is hidden and cannot be accessed by external functions.
 (A) OOP (B) SOP
 (C) POP (D) None of these
- 4) _____ is a selection control structure.
 (A) if (B) switch
 (C) if-else (D) All of the Above
- 5) _____ function is used to copy one string into another string.
 (A) strcpy() (B) stringcpy()
 (C) strcpry() (D) StrCopy()
- 6) For string manipulation _____ header file is included.
 (A) iostream.h (B) ctype.h
 (C) conio.h (D) string.h
- 7) Which of the following function allows multiple lines of input through the keyboard?
 (A) gets() (B) getline()
 (C) getch() (D) Both (B) and (C)
- 8) While defining the derived class, base class name is written on _____ side of : symbol.
 (A) Both (B) Any one
 (C) LHS (D) RHS
- 9) The correct function name for overloading the addition (+) operator is
 (A) operator+ (B) +operator
 (C) operator : + (D) operator(+)
- 10) Which one of the following operators can be overloaded?
 (A) sizeof (B) ::
 (C) .* (D) new

Q-2

Fill in the blanks/True-False.

(08)

- 1) _____ is basic run time entity in object-oriented system.
- 2) One base class & multiple derived classes means _____ inheritance
- 3) Constructor is special member function because its name is same as the _____ name.

- 4) Defining a function with same name but different types & no. of arguments is known as _____.
- 5) Bjarne Strastrup is the developer of C++.(True/False)
- 6) Public access specifier is default in class definition. (True/False)
- 7) By default, main () in C++ returns void. (True/False)
- 8) Certain overloaded operators can change the number of arguments they take. (True/False)

Q-3 Short Answer Questions. (Attempt any 10 out of 12)

(20)

- 1) Define polymorphism as concept of OOP.
- 2) What is object – oriented programming?
- 3) List out any four header file in C++.
- 4) Define constant and nameless objects.
- 5) Explain cout and cin in C++ with example.
- 6) Write general form of member function definition for outside the class definition.
- 7) Define virtual function. Also, state difference between virtual function & pure virtual function.
- 8) Discuss default arguments briefly.
- 9) What is a virtual base-class? What is the use of it?
- 10) List some of the rules for overloading operator.
- 11) What are input and output stream?
- 12) List out five operators that can be overloaded in C++.

Q-4 Long Answer Questions.(Attempt any 4 out of 8)

(32)

- 1) Describe the features of OOP in detail. Explain the structure of C++ by giving example.
- 2) Explain different data types available in C++.
- 3) How you can create arrays of objects in C++? Give an example and explain it.
- 4) Describe default, parameterized and copy constructor with example.
- 5) Explain the concept of friend function and inline functions with suitable example.
- 6) Explain multiple & multilevel inheritance with suitable example.
- 7) Explain binary operator overloading using member function and friend function with proper example.
- 8) What is file mode? Describe the various file mode operations available. Describe the various classes available for file operations.